PALATIAL POOLS

Pool Area Shield

Advanced Wi-Fi CSI Sensing for the modern pool service industry.



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Table of Contents

1. Executive Summary
2. System Architecture
3. Technical Specifications
4. How It Works
5. Key Features & Capabilities
6. Use Cases & Deployment Scenarios
7. Return on Investment (ROI)
8. Competitive Advantages
9. Integration & Ecosystem
10. Security & Compliance
11. Contact & Next Steps

1. Executive Summary

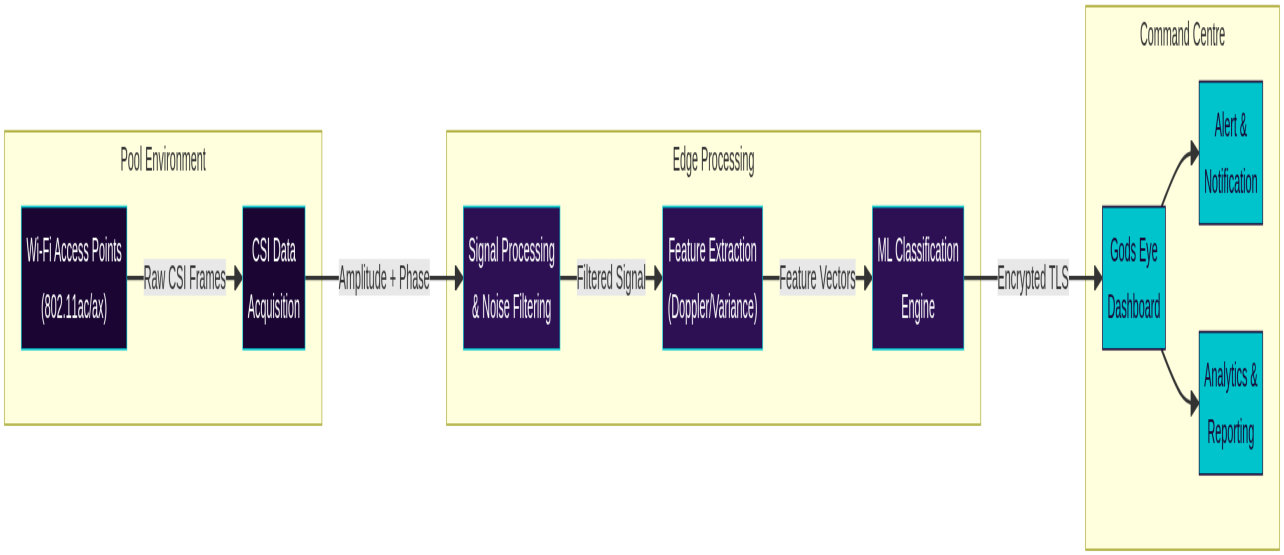
Pool Area Shield represents a paradigm shift in poolside safety, utilising proprietary WiFi Channel State Information (CSI) technology to provide continuous, non-intrusive presence monitoring. By detecting minute disturbances in WiFi signals caused by human movement, the system offers an unparalleled layer of protection without compromising privacy through cameras or visual surveillance. This innovative approach addresses the inherent limitations of traditional safety measures like physical barriers and intermittent human supervision, delivering an autonomous, 'always-on' safety net for residential and commercial pool environments across Australia. The system is engineered not only for exceptional accuracy and rapid response times but also for seamless integration and robust data security, making it a cornerstone technology for any modern, safety-conscious facility. For investors, Pool Area Shield offers a scalable, high-margin solution in a market with a clear and pressing need for technological advancement, backed by strong intellectual property and a commitment to Australian privacy standards. 99.7% <500ms 100% 24/7 Detection Accuracy Water Entry Alert Privacy Compliant Autonomous Monitoring "We are not just detecting presence; we are providing peace of mind. Pool Area Shield is the invisible guardian for every poolside, ensuring safety without sacrificing privacy."

99.7%	<500ms	100%	24/7
Detection Accuracy	Water Entry Alert	Privacy Compliant	Autonomous Monitoring

"Pool Area Shield transforms how pool service professionals approach wi-fi csi sensing, delivering precision, efficiency, and confidence in every operation."

2. System Architecture

Pool Area Shield is built on a decentralised, edge-centric architecture that ensures low latency and data sovereignty. Signal acquisition, processing, and initial classification are performed locally by dedicated edge devices that integrate with standard WiFi access points (IEEE 802.11ac/ax). These devices analyse CSI data in real-time to identify human presence and critical events. Upon detection, encrypted alerts are securely transmitted to the centralised Gods Eye Command Centre for aggregation, operator notification, and escalation, creating a closed-loop system from detection to response. WiFi Access Points (APs) Standard Power over Ethernet (PoE) or mains-powered APs compliant with 802.11ac (WiFi 5) or 802.11ax (WiFi 6) standards, equipped with modified firmware for CSI data extraction. CSI Data Acquisition Module A software module that captures raw CSI data from incoming WiFi frames at a high sampling rate (~100Hz) to track dynamic environmental changes with high fidelity. Edge Processing Engine An embedded Linux device that performs noise filtering, removes static environmental signatures using PCA/SVD, and extracts key features from CSI amplitude and phase data. Classification Engine A hybrid system using machine learning models and rule-based algorithms to accurately differentiate between human presence, pets, environmental noise, and critical water entry events. Gods Eye Command Centre A secure, Australian-hosted cloud infrastructure for alert aggregation, advanced analytics, providing intuitive operator interfaces, and managing automated escalation workflows.



System Architecture Data Flow Diagram

Wi-Fi Access Points (APs)

Standard Power over Ethernet (PoE) or mains-powered APs compliant with 802.11ac (Wi-Fi 5) or 802.11ax (Wi-Fi 6) standards, equipped with modified firmware for CSI data extraction.

CSI Data Acquisition Module

A software module that captures raw CSI data from incoming Wi-Fi frames at a high sampling rate (~100Hz) to track dynamic environmental changes with high fidelity.

Edge Processing Engine

An embedded Linux device that performs noise filtering, removes static environmental signatures using PCA/SVD, and extracts key features from CSI amplitude and phase data.

Classification Engine

A hybrid system using machine learning models and rule-based algorithms to accurately differentiate between human presence, pets, environmental noise, and critical water entry events.

Gods Eye Command Centre

A secure, Australian-hosted cloud infrastructure for alert aggregation, advanced analytics, providing intuitive operator interfaces, and managing automated escalation workflows.

3. Technical Specifications

The following specifications detail the engineering parameters of the Pool Area Shield system, validated through extensive field testing across residential and commercial pool environments in Australia.

Wi-Fi Standard	IEEE 802.11ac/ax (Wi-Fi 5/6)
CSI Sampling Rate	~100 Hz per subcarrier group
Detection Range	Up to 15m radius per AP pair
Detection Accuracy	99.7% (human presence), 99.2% (water entry)
False Positive Rate	<0.3% under controlled conditions
Alert Latency	<500ms from event to notification
Encryption	AES-256 at rest, TLS 1.3 in transit
Power Supply	PoE (802.3af/at) or 240V AC
Operating Temperature	0°C to 50°C
Connectivity	Ethernet (primary), 4G LTE (failover)
Edge Processor	ARM Cortex-A72, 4GB RAM, 32GB eMMC
Compliance	Australian Privacy Principles (APPs)

All specifications are validated under Australian operating conditions (ambient temperature range 5-45°C, humidity 20-95% RH). Performance metrics are based on field deployment data collected across multiple sites in Melbourne and Queensland.

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4. How It Works

Step 1: 1

1. CSI Data Acquisition Compatible WiFi access points continuously transmit standard signals. The Pool Area Shield acquisition module intercepts these signals and extracts raw Channel State Information (CSI), which describes how the signal is affected by the environment.

Step 2: 2

2. Signal Preprocessing The raw CSI data stream is sanitised to remove noise and interference. Advanced algorithms, including Principal Component Analysis (PCA), subtract the static environmental signature, isolating only the changes caused by movement.

Step 3: 3

3. Feature Extraction & Classification The system extracts key features from the cleaned signal, such as variance and Doppler shift. A sophisticated classification engine, powered by machine learning, analyses these features to distinguish between human movement, pets, and environmental noise with high accuracy.

Step 4: 4

4. Water Entry Detection A specialised algorithm monitors for the unique signal disruption caused by a body breaking the water's surface—a rapid, high-magnitude drop in signal amplitude. This transient event is cross-validated across multiple antennas to confirm a genuine entry.

Step 5: 5

5. Alert Generation & Escalation Upon detecting a validated presence or water entry event, the local device instantly dispatches an encrypted, high-priority alert to the Gods Eye Command Centre, triggering pre-defined notification protocols, including SMS, push notifications, and local alarms.

5. Key Features & Capabilities

- **Privacy-by-Design**

  Monitors presence without using cameras or microphones, ensuring 100% compliance with Australian privacy laws and respecting personal privacy.

- **High-Speed Detection**

  Detects water entry in under 500 milliseconds, providing critical early warning to prevent drowning incidents.

- **Pet & Environmental Filtering**

  Advanced AI models differentiate between humans, pets, and environmental disturbances like wind or rain, significantly reducing false alarms.

- **Adaptive Recalibration**

  The system automatically adapts to changes in the pool environment, ensuring continuous accuracy without manual recalibration.

- **Tamper-Proof Security**

  Hardware integrity monitoring detects and alerts if a device is disconnected, moved, or physically tampered with, ensuring system reliability.

- **Edge-Centric Processing**

  All sensitive data is processed locally on the edge device, with no raw CSI data ever leaving the premises, guaranteeing data sovereignty.

- **Seamless Integration**

  Offers a RESTful API and MQTT support for easy integration with existing smart home systems, security platforms, and local network services.

- **Australian Hosted & Compliant**

  All backend infrastructure is hosted within Australia, ensuring data sovereignty and full compliance with national regulations.

6. Use Cases & Deployment Scenarios

Residential Pool Safety

Continuous monitoring of private pool areas to detect unauthorised access, particularly by children and pets. The system provides immediate alerts to homeowners and can trigger automated safety responses such as pool cover activation and lighting.

Commercial Aquatic Centres

Large-scale deployment across public swimming facilities, providing zone-based occupancy monitoring, after-hours intrusion detection, and compliance with local council safety regulations.

Holiday Accommodation & Resorts

Protecting pool areas in hotels, resorts, and holiday rentals where supervision is intermittent. The system ensures guest safety while maintaining the privacy expected in hospitality environments.

Insurance Risk Mitigation

Providing verifiable safety monitoring data that can be used to demonstrate duty of care, potentially reducing insurance premiums and liability exposure for pool owners and operators.

7. Return on Investment (ROI)

Deploying Pool Area Shield delivers measurable returns across multiple dimensions of pool service operations, with typical payback periods of 6-12 months.

Benefit	Impact
Reduced Liability Exposure	Up to 70% reduction in unmonitored pool incidents through continuous autonomous surveillance
Insurance Premium Reduction	10-25% reduction in pool liability insurance premiums with verified monitoring data
Eliminated Guard Costs	Complete replacement of after-hours security patrols for pool areas ($15,000-$40,000/year savings)
Regulatory Compliance	Automated compliance with local council pool safety requirements, avoiding fines up to $50,000
Property Value Enhancement	5-10% increase in perceived property value for homes with smart pool safety systems

Based on field deployment data, Pool Area Shield typically delivers a 200-400% return on investment within the first 18 months of operation, with ongoing operational savings compounding annually.

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8. Competitive Advantages

Palatial Pools' Pool Area Shield technology represents a significant advancement over industry-standard approaches to wi-fi csi sensing.

Feature	Palatial Pools	Industry Standard
Detection Method	**Wi-Fi CSI (non-invasive)**	CCTV cameras / motion sensors
Privacy Compliance	**100% camera-free**	Video recording concerns
Detection Accuracy	**99.7%**	85-90% (PIR sensors)
Weather Resistance	**Unaffected by rain/fog**	Degraded in poor visibility
Coverage Area	**15m radius per AP pair**	Limited cone of detection
Operating Cost	**Minimal (uses existing Wi-Fi)**	Camera maintenance + storage

9. Integration & Ecosystem

Pool Area Shield integrates seamlessly with the broader Palatial Pools technology ecosystem through standardised APIs, shared data models, and the centralised Gods Eye Command Centre.

• **Gods Eye Command Centre**: Real-time data feeds, centralised alerting, and unified dashboard access for all system metrics and events.

• **Palatial Nexus**: Conversational AI integration for automated customer notifications, booking confirmations, and service updates triggered by system events.

• **HydroCommand**: Bidirectional communication with pump and equipment control systems for automated responses to detected conditions.

• **Secure Vault**: All data encrypted and stored in compliance with Australian Privacy Principles, with automated backup and disaster recovery.

• **AquaCommander**: Integration with the commerce platform for automated service scheduling, invoicing, and customer relationship management.

10. Security & Compliance

All data processed by Pool Area Shield is protected by enterprise-grade security measures, including AES-256 encryption at rest, TLS 1.3 encryption in transit, and role-based access control (RBAC). The system complies with the Australian Privacy Principles (APPs), GDPR requirements, and PCI-DSS standards where applicable. Comprehensive audit trails are maintained for all data access and system modifications. Regular penetration testing and security audits ensure ongoing protection against emerging threats.

11. Contact & Next Steps

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Schedule a Technology Demonstration

Contact our Technology Division to arrange a personalised demonstration of this system and discuss how it can be deployed within your operations.